Ch 7.2-7.4: Step Functions, Basis Functions, Start Splines

Lecture 22 - CMSE 381

Prof. Elizabeth Munch

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Weds, Oct 30, 2024

Announcements

Last time:

- 7.1 Polynomial regression
- 7.2 Step functions

This lecture:

- 7.2 Step functions
- 7.3 Basis functions
- 7.4 Regression Splines (Finish next lecture)

Announcements:

- HW #6 Due Sunday
- Projects

Section 1

Last time

Polynomial regression

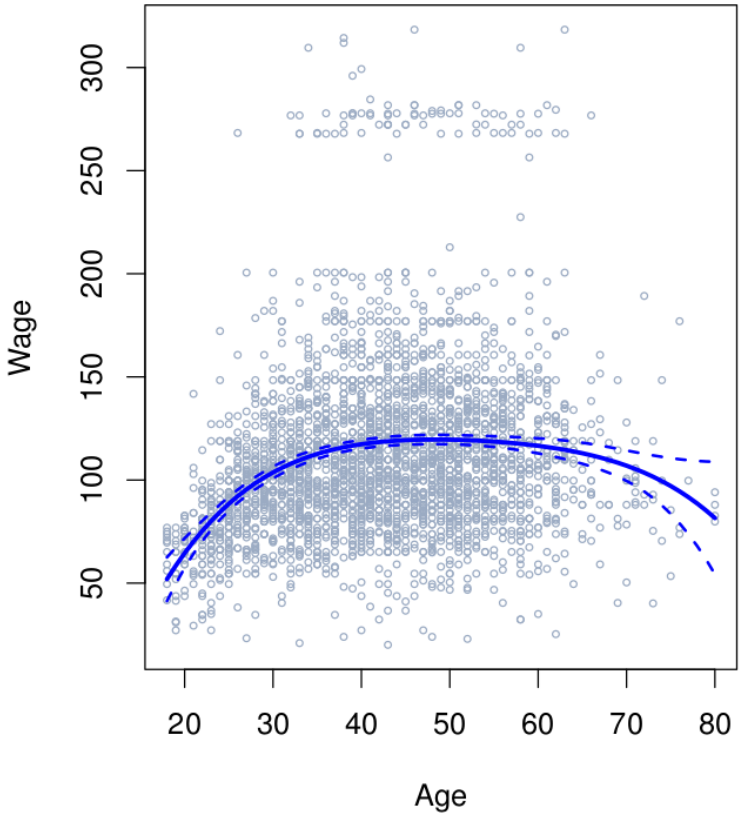
Replace linear model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

with

$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i$$

Example with wage data



$$-184.1542 + 21.24552 * age + -0.56386 * age^2 + 0.00681 * age^3 + -3e - 05 * age^4$$

Step functions

cuts $c_1 < c_2 < c_3 \ldots < c_K$

- $\mathrm{I}(X < c)$
- $\mathrm{I}(c_1 \leq X < c_2)$
- $\mathrm{I}(c \leq X)$

$$
\begin{aligned}
C_0(X) &= I(X < c_1), \\
C_1(X) &= I(c_1 \leq X < c_2), \\
C_2(X) &= I(c_2 \leq X < c_3), \\
&\vdots \\
C_{K-1}(X) &= I(c_{K-1} \leq X < c_K), \\
C_K(X) &= I(c_K \leq X),
\end{aligned}
$$

1 if X in set
0 else

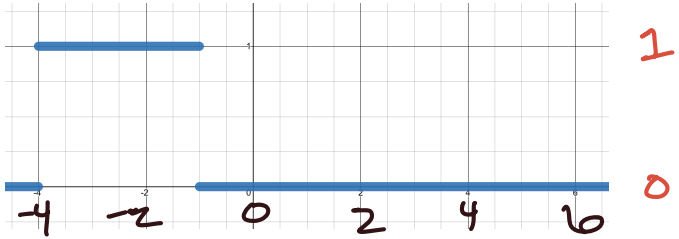
Learned model:

$$y_i = \beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i) + \varepsilon_i$$

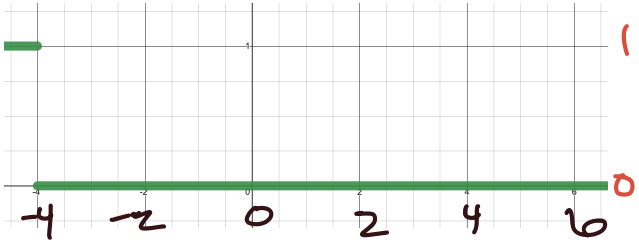
Example: Cut points at -4, -1, 3, 6

$c_1 \quad c_2 \quad c_3 \quad c_4$

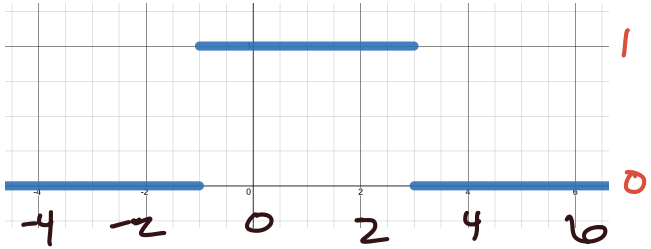
$C_1(X)$



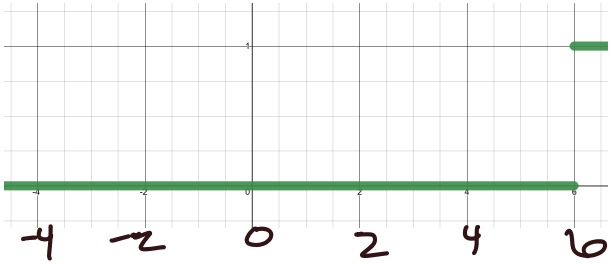
$C_0(X)$



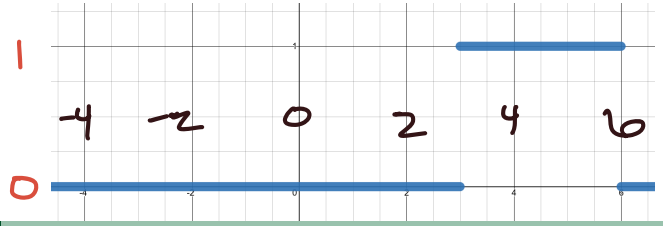
$C_2(X)$



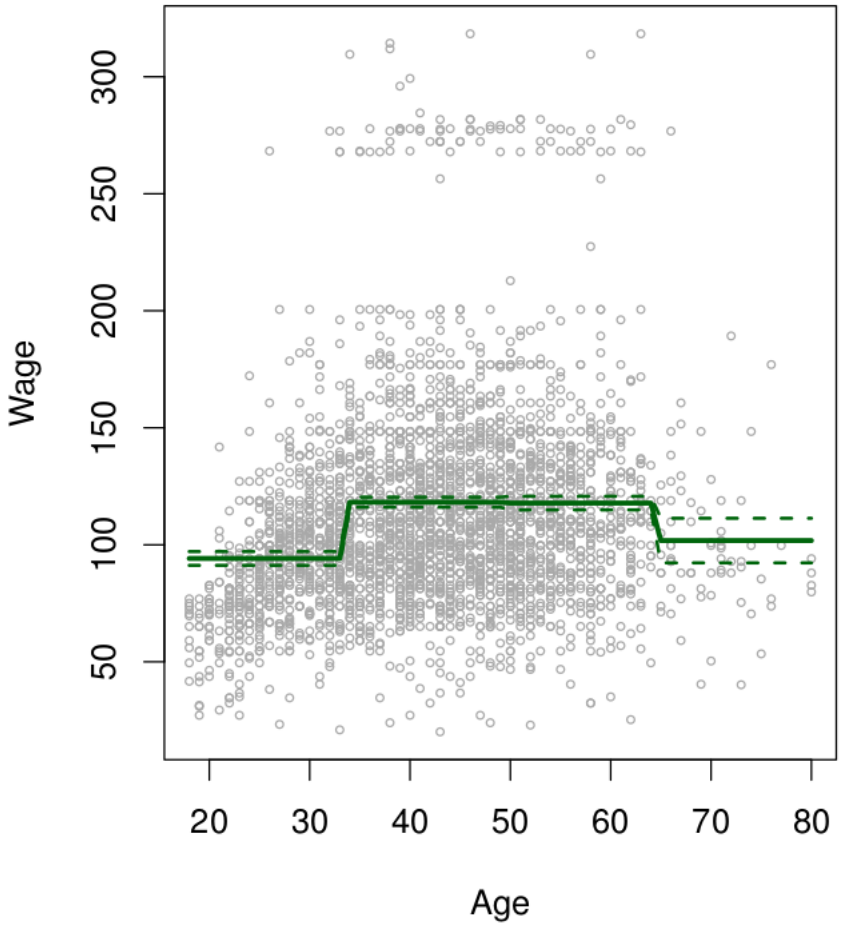
$C_4(X)$



$C_3(X)$



Step function example

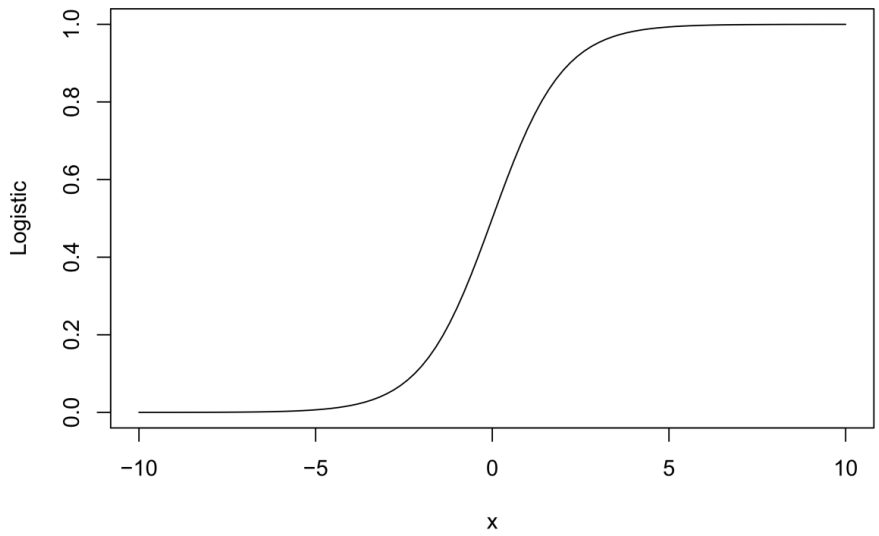


Section 2

Classification versions

Remember logisitic regression?

$$y = \frac{e^x}{1 + e^x}$$



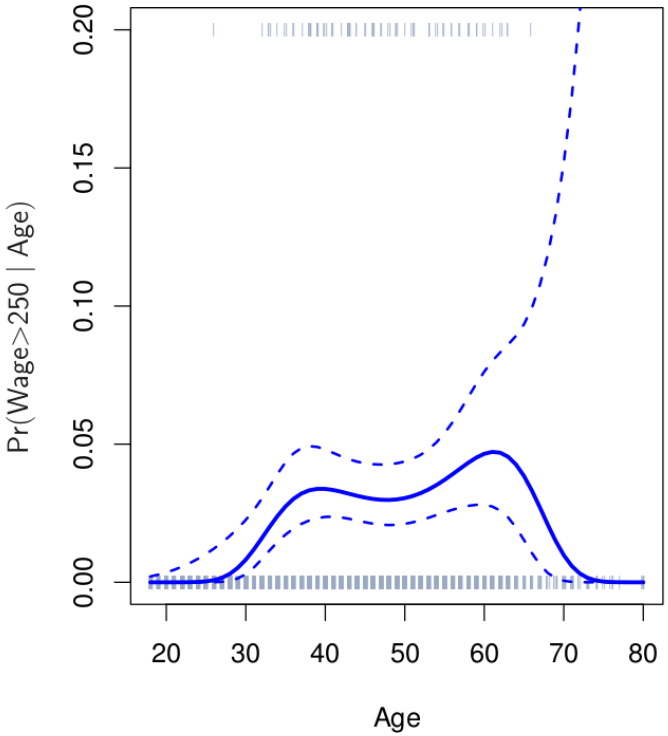
$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

Multiple features:

$$p(X) = \frac{e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p}}$$

Classification version: Polynomial regression

$$\Pr(y_i > 250 \mid x_i) = \frac{\exp(\beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d)}{1 + \exp(\beta_0 + \beta_1 x_i + \cdots + \beta_d x_i^d)}$$



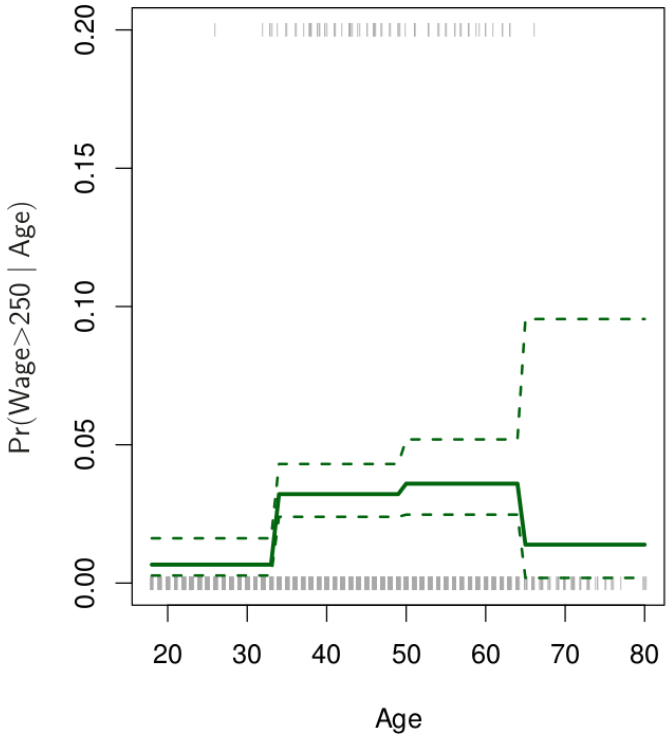
Classification problem

Predict if person makes ≥ $250 K

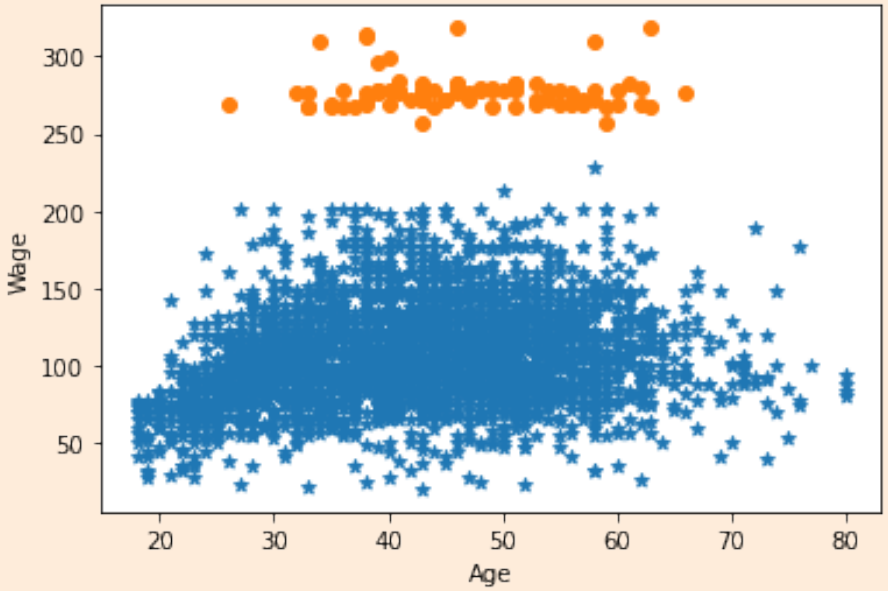
Dashed line gives pointwise CI

Classification version: Step functions

$$\Pr(y_i > 250 \mid x_i) = \frac{\exp(\beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i))}{1 + \exp(\beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \cdots + \beta_K C_K(x_i))}$$



Coding bit: classification version



A few more comments on step functions

- Gives the chance to break up the domain, avoid forcing global structure
- Need to make decisions about the c_i. A bit arbitrary unless your data has natural breakpoints.
- Popular in biostats and epidemiology

Section 3

Basis functions

Basis Functions Setup

Polynomial and piecewise-constant regression models are special cases of a *basis function* approach.

$$y_i = \beta_0 + \beta_1 b_1(x_i) + \beta_2 b_2(x_i) + \cdots + \beta_K b_K(x_i) + \varepsilon_i$$

- Pick collection of functions $\{b_1(x), __, b_k(x)\}$

  $x \quad x^2 \ldots x^k$

  $c_1(x) \quad c_2(x) \ldots c_k(x)$

- Least squares $\Rightarrow$ figure out coef.
- Figure out the function from the learned coef.

Section 4

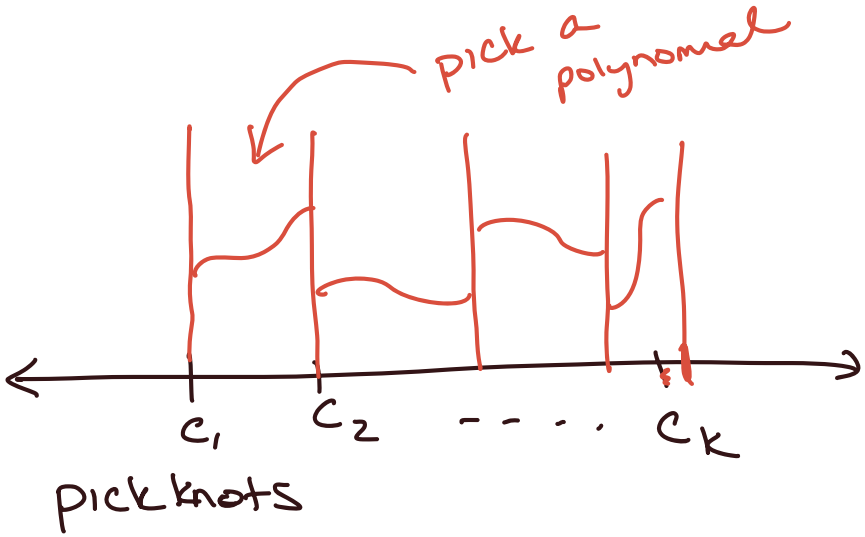
Regression Splines

Piecewise polynomials

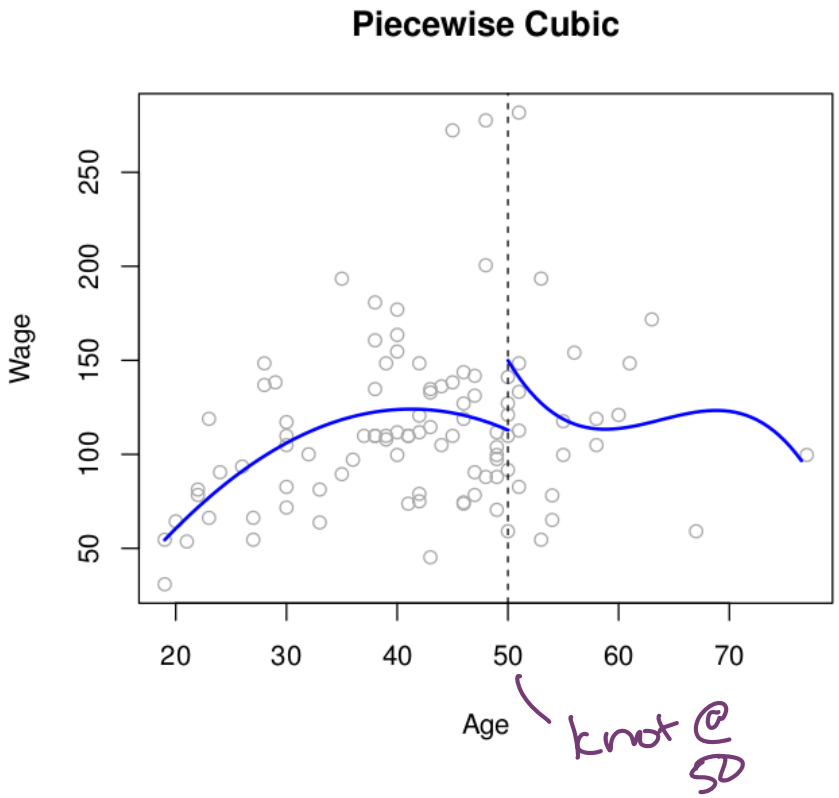
- Fit a polynomial regression

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_d x_i^d + \varepsilon_i$$

- Let the β_i's be different at different locations of the range.



Example of piecewise polynomial



Example:

$$y_i = \begin{cases} \beta_{01} + \beta_{11}x_i + \beta_{21}x_i^2 + \beta_{31}x_i^3 + \epsilon_i & \text{if } x_i < c \\ \beta_{02} + \beta_{12}x_i + \beta_{22}x_i^2 + \beta_{32}x_i^3 + \epsilon_i & \text{if } x_i \geq c. \end{cases}$$

Decide on 8 #'s → piecewise cubic here

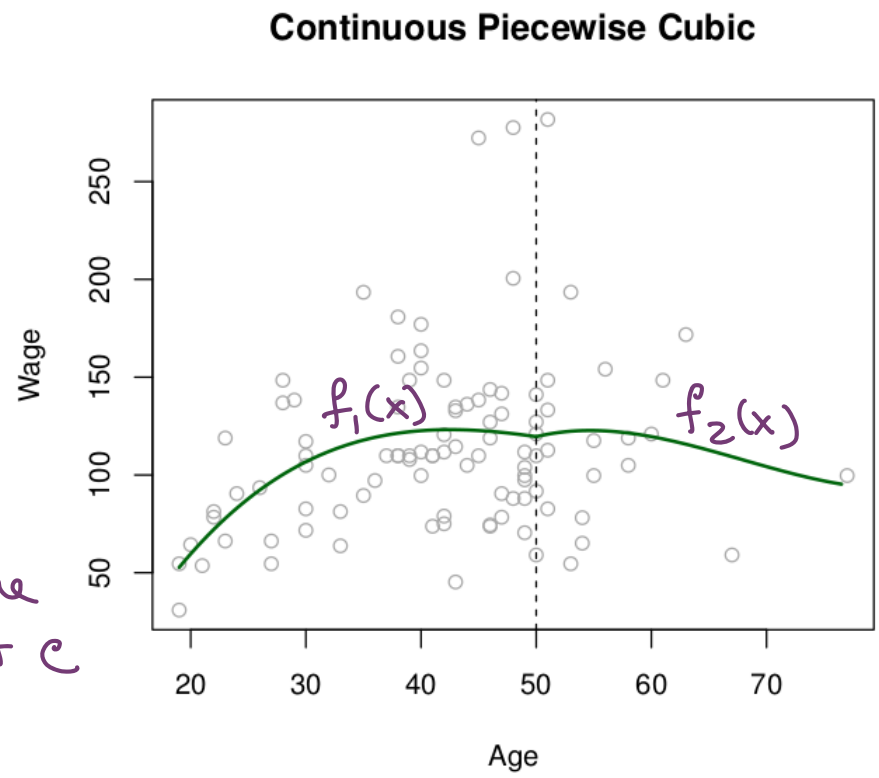
Prob: This is discontinuous

The fix

- Fit piecewise polynomial
- Require continuity at knots

$f_1(c) = f_2(c)$ at the knot c

Prob: not smooth



The better fix: Cubic splines

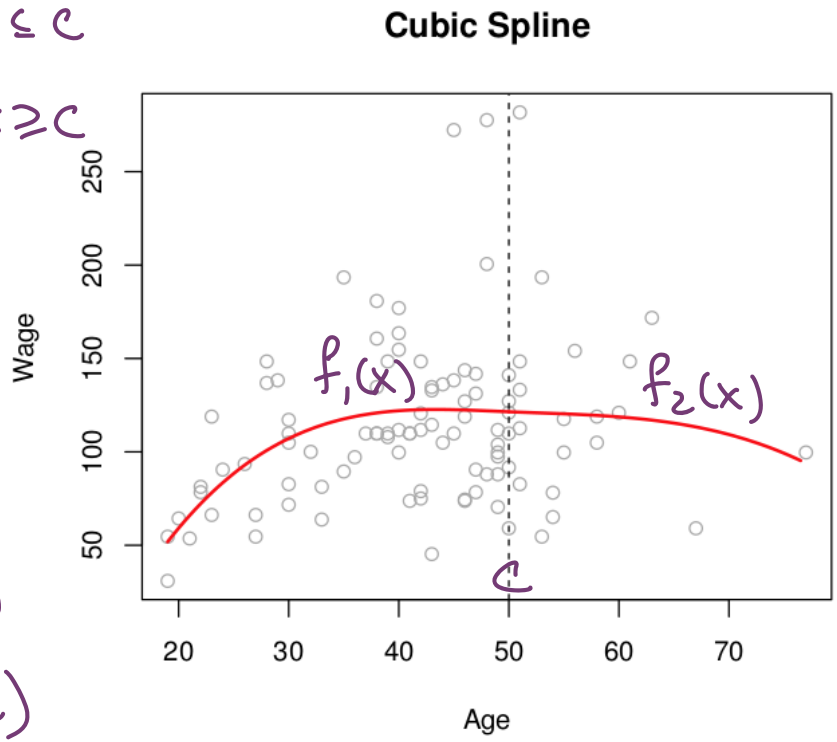
$$f(x) = \begin{cases} f_1(x) & \text{if } x \le c \\ f_2(x) & \text{if } x \ge c \end{cases}$$

Cubic spline:

- Fit piecewise polynomial
- Require continuity at knots $f_1(c) = f_2(c)$
- Require the first and second derivatives to be continuous at knots

$$f_1'(c) = f_2'(c)$$

$$f_1''(c) = f_2''(c)$$



One knot

- 8 coef to work w/ 8 degrees of freedom
- Requirements mean $\Rightarrow 8-3 = 5$ degrees of freedom.

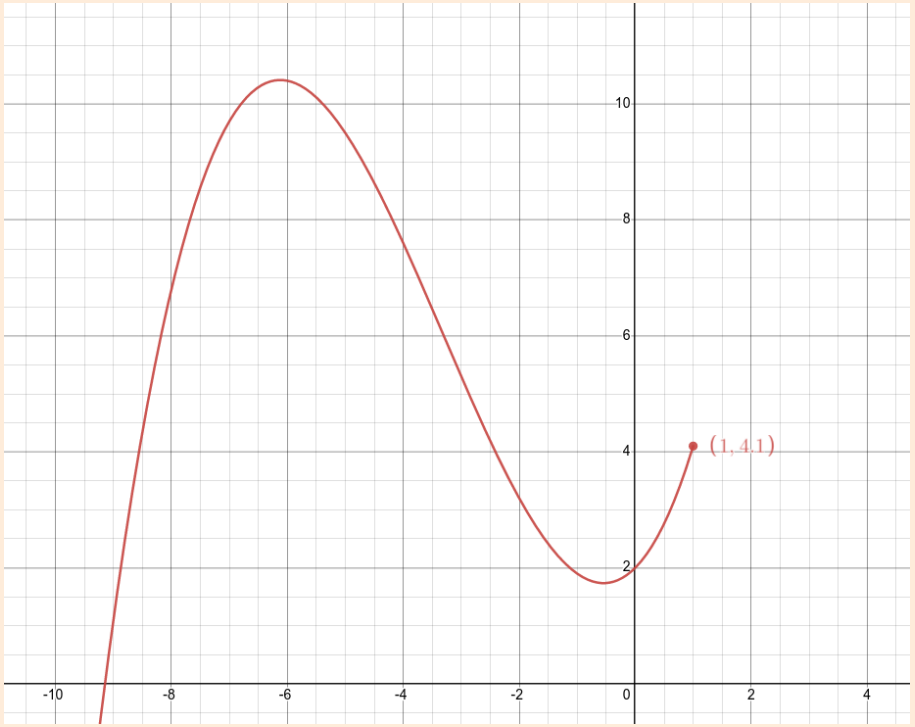
K knots: $4+K$ degrees of freedom.

Example

We have the following piecewise cubic polynomial:

$$f(x) = \begin{cases} 2 + x + x^2 + 0.1x^3 & x \leq 1 \\ b_0 + b_1 x + b_2 x^2 - x^3 & x > 1 \end{cases}$$

What are b_1, b_1, and b_2 to make this a cubic spline?



Check your answers: desmos.com/calculator/ns4tr7mw0n

More space for work

$$f(x) = \begin{cases} 2 + x + x^2 + 0.1x^3 & x \le 1 \\ b_0 + b_1 x + b_2 x^2 - x^3 & x > 1 \end{cases}$$

@ 1

$f_1(x) = 2 + x + x^2 + 0.1x^3$ | $2+1+1+0.1 = 4.1$ | $f_2(x) = b_0 + b_1 x + b_2 x^2 - x^3$

$f_1'(x) = 1 + 2x + 0.3x^2$ | $1+2+0.3 = 3.3$ | $f_2'(x) = b_1 + 2b_2 x - 3x^2$

$f_1''(x) = 2 + 0.6x$ | 2.6 | $f_2''(x) = 2b_2 - 6x$

$2b_2 - 6 = 2.6$

$2b_2 = 8.6$

$\boxed{b_2 = 4.3}$

$b_1 + 2b_2 - 3 = 3.3$

$b_1 + 2(4.3) - 3 = 3.3$

$b_1 + 8.6 - 3 = 3.3$

$b_1 + 5.6 = 3.3$

$\boxed{b_1 = -2.3}$

$b_0 + (-2.3) + 4.3 - 1 = 4.1$

$b_0 + 1 = 4.1$

$\boxed{b_0 = 3.1}$

Next time

Lec #	Date			Reading	HW
21	Mon	10/28	Polynomial & Step Functions	7.1,7.2	
22	Wed	10/30	Step Functions; Basis functions; Start Splines	7.2 - 7.4	
23	Fri	11/1	Regression Splines	7.4	HW #7 Due Sun 11/3
24	Mon	11/4	Decision Trees	8.1	
25	Wed	11/6	Class Cancelled (Dr Munch out of town)		
26	Fri	11/8	Random Forests	8.2.1, 8.2.2	HW #8 Due Sun 11/10
27	Mon	11/11	Maximal Margin Classifier	9.1	
28	Wed	11/13	SVC	9.2	
29	Fri	11/15	SVM	9.3, 9.4	HW #9 Due Sun 11/17
30	Mon	11/18	Single layer NN	10.1	
31	Wed	11/20	Multi Layer NN	10.2	
32	Fri	11/22	CNN	10.3	HW #10 Due Sun 11/24
33	Mon	11/25	TBD: Unsupervised learning/clustering	12.1, 12.4?	
	Wed	11/27	Virtual: Project office hours		